Supervised Learning for Medellin Land Cover Classification

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Abstract

Urban green infrastructure plays a critical role in climate change mitigation and urban heat island reduction. This study evaluates the effectiveness of Medellín's "Corredores Verdes" initiative through multi-temporal analysis of Sentinel-2 and Landsat 8 imagery. A binary classification approach (vegetation vs. non-vegetation) was implemented using Random Forest (RF) and Support Vector Machine (SVM) classifiers, trained on 679 spatially-distributed samples with 10 spectral, topographic, and textural features. Both models achieved high performance (RF: AUC=0.989; SVM: AUC=0.997), with SVM demonstrating superior accuracy (98.2%). Change detection between 2016 and 2025 revealed a 46.2% increase in citywide vegetation and 110.6% in corridor areas, accompanied by temperature reductions of up to 5°C. Spatial validation on Bogota confirmed model transferability (93.5% accuracy). The methodology provides a robust, replicable framework for monitoring urban reforestation programs and demonstrates the tangible benefits of green infrastructure in tropical cities.

Keywords: Earth Observation, Random Forest, Support Vector Machine, Urban Greening, Change Detection, Heat Island Mitigation

1 Introduction

Medellín, Colombia, has implemented the "Corredores Verdes" program since 2016 to reduce urban overheating through targeted reforestation along major streets. This initiative offers a valuable case study for assessing the impact of urban greening using remote sensing.

Advances in Earth Observation and machine learning now enable cost-effective monitoring of land-cover change. In particular, Sentinel-2 multispectral imagery and supervised classifiers such as Random Forest and Support Vector Machines have proven effective for vegetation mapping in complex urban environments.

2 Materials and Methods

All analyses were performed in Python using data retrieved from Google Earth Engine (GEE), including Sentinel-2, Landsat-8, and a digital elevation model. The Medellín municipal boundary (FAO GAUL 2015) was used as area of interest.

To represent comparable seasonal conditions, summer images (November–March) were selected for the pre-policy period (2015–2016), post-policy period (2024–2025). Cloud masking and a median composite were applied to generate the final images.

A total of 350 points per class (non-vegetation/vegetation) were manually sampled from the 2024–2025 Sentinel-2 image. Fifteen features were extracted for each point:

- **spectral bands**: B2, B3, B4, B8, B11, B12
- **indices**: NDVI, EVI, SAVI, NDBI
- **topography**: elevation, slope, aspect
- **textures**: NIR contrast, entropy (3×3 window)

A dedicated "corridors" layer was created from municipal data [1] by buffering relevant streets by 150 m.

2.1 Data Preparation

Before training the models data cleaning and preparation were performed: (i) removal of null values; (ii) outlier filtering (z-score), yielding 679 samples (50.8% class 0, 49.2% class 1); (iii) correlation analysis to remove redundant bands (Figure 3); (iv) sequential feature selection, identifying [B2, B11, NDBI, elevation, slope, aspect, contrast, entropy] for RF (accuracy 0.975) and [B8, NDVI] for SVM (accuracy 0.958); (v) spatial distribution filtering (grid, k-means, and minimum-distance). The last method yielded the best balance, producing 400 training points (178 class 0, 222 class 1); (vi) normalization for SVM.

2.2 Training and validation

The training of the model was performed with the best features found with spatial distribution analysis and a **grid research** was conducted for both models in order to find the best model parameters.

A cross validation is performed, for a most robust assessment of the models performances, using 5 folds and the model performance was evaluated using confusion

matrices, overall accuracy, precision, recall, F1-score, ROC curves, and AUC values.

2.3 Test

The best-performing model was further evaluated through a robustness test by applying the model to the urban area of Bogota, Colombia, using 150 points sampled for each class. This ensured consistency of performance across a similar but geographically distinct urban area. Before testing the same data preparation was applied to the test set.

2.4 Change detection

To verify effective changes in land cover and temperature, the years 2016 and 2025 were compared during the summer season to avoid the rainy period and better monitor vegetation. Two types of comparison were made: (i) **thermal changes** and (ii) **land cover changes**.

(i) For the thermal analysis, the thermal infrared band of Landsat 8 was used to derive the land surface temperature for 2016 and 2025 using equation (3), where ST_B10 is the digital number of the image for that band, multiplied by the multiplication factor and added to the additive factor [2].

$$T_K = (\text{ST_B10} \times 0.00341802) + 149.0 \qquad (1)$$

The temperature was then converted from Kelvin to Celsius degrees by subtracting 273.15, and the difference between 2016 and 2025 was computed to assess thermal variations.

(ii) The changes in land cover were computed by applying the classifier to the images of the area of interest (AOI) and calculating the difference between the 2025 and 2016 vegetation areas.

2.4.1 Green corridors

A more specific analysis was carried out in the buffer zone around the reforested streets. Differences in land surface temperature, and vegetation area were computed to evaluate the effectiveness of the "Corredores Verdes" initiative.

3 Results

3.1 Classifier performance

The performance of Random Forest (RF) and Support Vector Machine (SVM) was evaluated using the metrics summarized in Table1. An overview of the performance is given by the confusion matrices below:

$$RF = \begin{bmatrix} 171 & 7 \\ 5 & 217 \end{bmatrix} \quad SVM = \begin{bmatrix} 171 & 7 \\ 4 & 218 \end{bmatrix}$$

Both models perform very well in classify both classes. SVM performs slightly better but we can say that they are both valid. Both classifiers achieved high accuracy, with SVM (C = 1, gamma = scale, kernel = linear) slightly outperforming RF. The confusion matrices reveal balanced error distribution: RF misclassified 12 samples (5 vegetation as non-vegetation, 7 vice versa), while SVM misclassified 11 samples (4+7). The high recall values (>0.96) indicate minimal false negatives, critical for vegetation monitoring applications

Table 1: Performance metrics of the classifiers.

Metric	RF	SVM
Accuracy	0.965	0.982
Precision	0.965	0.973
Recall	0.966	0.973
F1-score	0.965	0.972
AUC	0.989	0.997

Figures 1 and 2 show the ROC curves for both classifiers which confirm model robustness, with AUC values exceeding 0.98 for both classifiers. The near-perfect discrimination (AUC_SVM=0.997) suggests excellent separability between non-vegetation and vegetated classes using the selected feature sets. Based on these results, SVM was selected for subsequent change detection analysis.

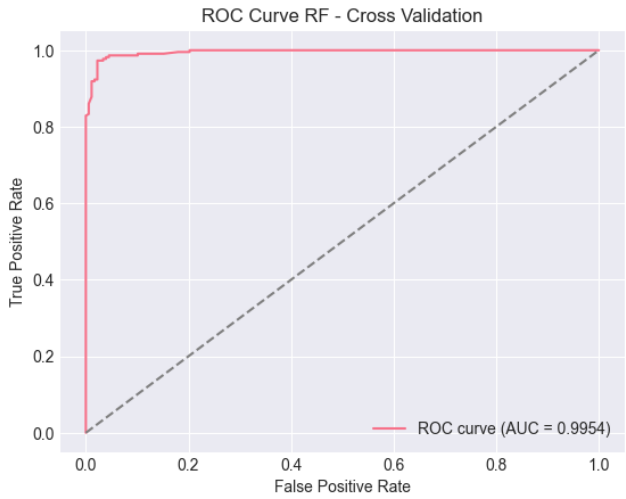

Figure 1: RF ROC curve

3.2 Model robustness test

The spatial test on Bogota (Table 2) also demonstrated high accuracy (94.0%). The modest accuracy reduction (4.7% compared to Medellín test set) is attributable to inter-city differences in urban morphology and vegetation types. Nevertheless, the >93% accuracy confirms the methodology's robustness and potential for regional-scale application across similar cities.

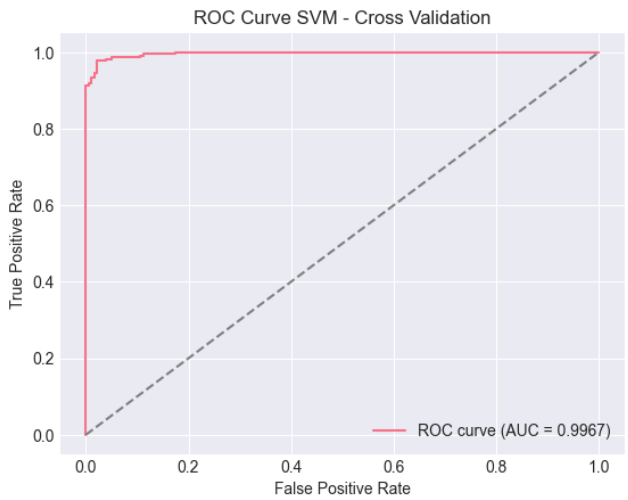

Figure 2: SVM ROC curve

Table 2: Performance of SVM on Bogota spatial test dataset.

Metric	**Value**
Accuracy	0.935
Precision	0.937
Recall	0.935
F1-score	0.935

3.3 Change detection

Table 3 reports the area covered by vegetation and non-vegetation surfaces in 2016 and 2025 (figures 4 and 5). Citywide vegetation increased by 86.3 km^2, representing a 46.2% expansion. Notably, the Corredores Verdes zones exhibited a 110.6% increase — more than doubling their green cover — providing compelling evidence of the intervention's success. (Figures 6).

Table 3: Change in vegetation class area (km^2).

	2016	2025	Change (%)
Entire municipality	186.7	273.0	+46.2%
Corredores Verdes	11.8	24.9	+111%

3.4 Thermal analysis

The land surface temperature (LST) is shown in Figure 7. Citywide average summer temperature decreased by 4.8°C, while corridor zones experienced 5.0°C cooling. The higher initial temperatures in corridors (35.4°C in 2016) reflect the thermal signature of asphalt dominance, which were subsequently mitigated through tree canopy establishment.

Table 4 reports the average LST values in 2016 and 2025. The results suggest that reforestation contributed to mitigating urban overheating, especially in the hottest areas represented by the principal roads, i.e., the area of green corridors.

Table 4: Average land surface temperature (°C).

	2016	2025	difference
Entire city	29.2	24.4	-4.8
Corridors	35.4	30.4	-5

4 Discussion

The results highlight how combining Sentinel-2 multispectral data with machine learning classifiers can effectively capture vegetation changes associated with Medellín's urban reforestation program. While both Random Forest (RF) and Support Vector Machine (SVM) performed well, the slightly higher accuracy of SVM suggests a better ability to separate vegetated from non-vegetated surfaces in dense urban settings.

The increase in vegetation along the "Corredores Verdes" and the corresponding reduction in the temperature of the land surface support previous evidence that linear green infrastructures improve microclimatic regulation.

Despite these encouraging results, several limitations should be noted. The binary classification simplifies the heterogeneity of urban land cover; medium-resolution imagery may overlook small patches of vegetation; and the temporal analysis, limited to two years, does not capture seasonal or interannual variability. Future research could explore multi-class schemes, incorporate higher-resolution optical or LiDAR data, and extend the analysis to different cities and climate contexts.

5 Conclusions

Sentinel-2 imagery combined with machine learning provides an efficient and scalable approach for monitoring urban reforestation. Between 2016 and 2025, Medellín experienced a clear expansion of vegetation cover—particularly along the "Corredores Verdes"—accompanied by a measurable reduction in land surface temperature.

These findings underline the role of targeted greening strategies in mitigating urban heat. Expanding the method with higher-resolution data and more detailed land-cover classes would further enhance its capacity to evaluate the ecological impact of urban reforestation initiatives.

Images

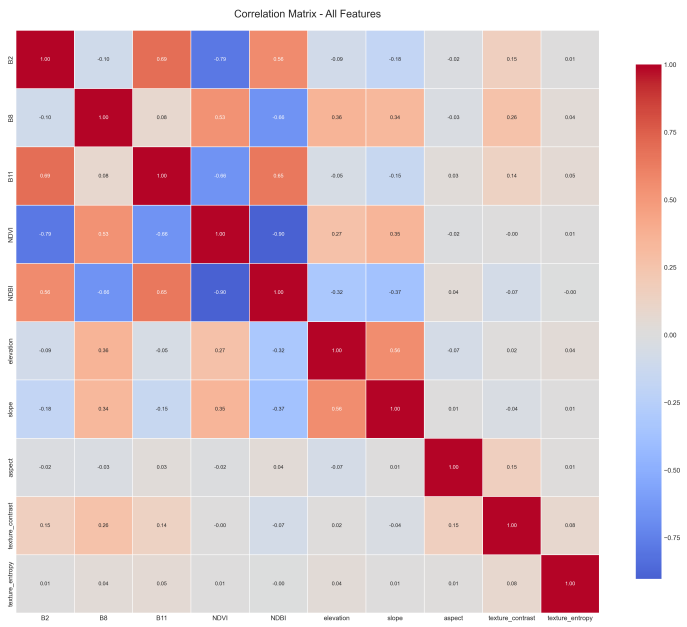

Figure 3: Features correlation heatmap

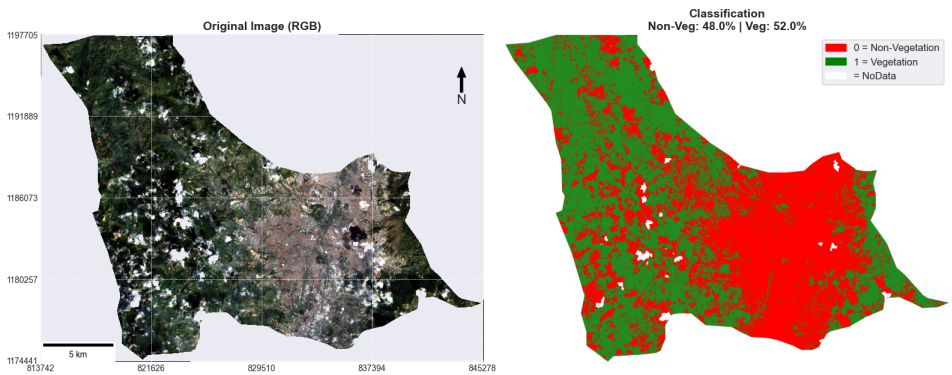

Figure 4: Land cover classification map of Medellín for 2016.

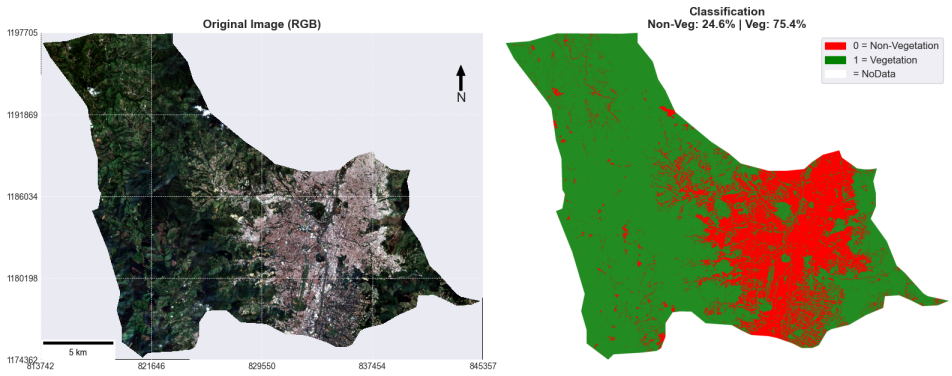

Figure 5: Land cover classification map for 2025.

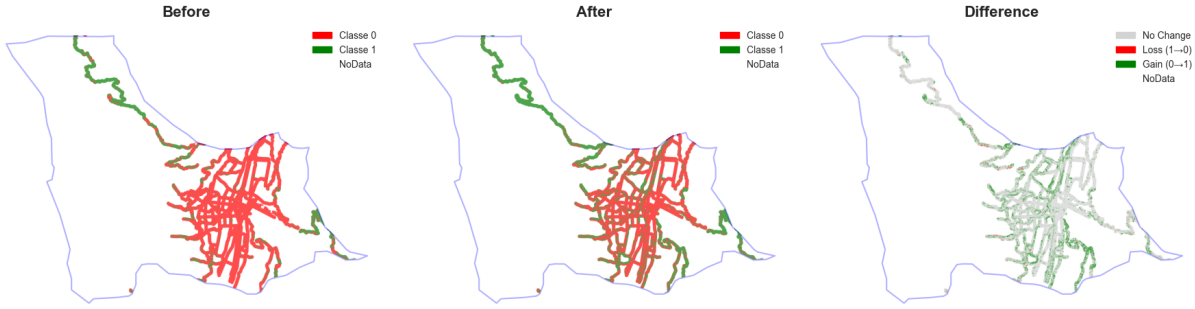

Figure 6: Land cover classification of green corridors.

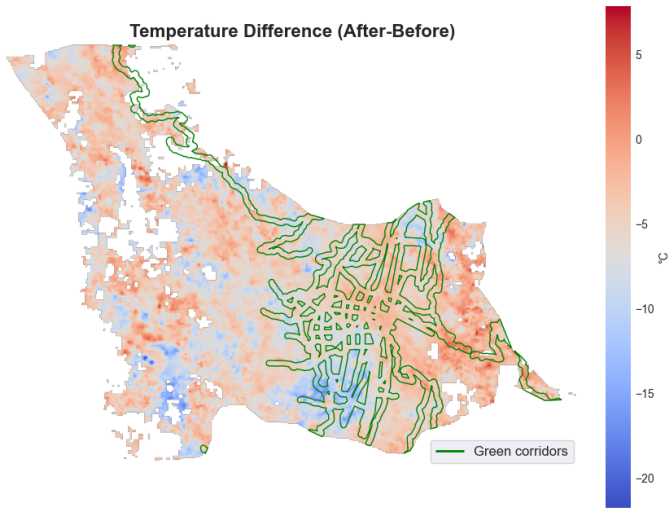

Figure 7: LST difference map of Medellin.

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